



In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No. : CE G614
Course Title : PRESTRESSED CONCRETE STRUCTURES
Instructor In-charge : Prof. Subhasis Pradhan

1. Course Description

Introduction to basic concepts of prestressing; prestressing systems; analysis of prestress; losses in prestress; deflection; design of anchorage zone, design of prestressed concrete girders.

2. Scope and Objectives

The universal use of prestressed concrete as a versatile construction material is well established due to its suitability in multifarious types of structural applications. The application of prestressed concrete is increasing with the development of technology. Prestressing is the application of an initial load on a structure to enable it to counteract the stresses arising from subsequent loads during its service period. In this course, the students will be introduced to prestressed concrete and its applications.

The course would cover important topics of prestressed concrete, including the design of structural elements based on the limit state philosophy of design. It would partially fulfill the requirements of practicing structural design engineers.

The course integrates science and engineering principles to design prestressed concrete members and structural systems. The application of scientific and engineering knowledge would be demonstrated in solving engineering problems associated with the design of precast prestressed building members. This course would be quite helpful for students who want to join the structural design or construction industry. At the end of the coursework, students will be able to determine the behavior of prestressed concrete structures.

2. Course Learning Outcomes

Upon successful completion of this course, students will be able to understand the fundamental concepts of prestressing and the different systems used in prestressed concrete construction. They will gain the ability to –

- Understand the basic principles of prestressing and identify various prestressing systems including pre-tensioning and post-tensioning methods.



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- Analyze the behaviour of prestressed concrete members under axial loads and bending, including the use of pressure/thrust lines and load balancing concepts.
- Calculate different types of prestress losses and understand their causes and effects on structural performance.
- Evaluate the flexural and shear strength of prestressed concrete members and design appropriate reinforcements.
- Analyze the transfer of prestress in pre-tensioned members and anchorage zone stresses in post-tensioned members.
- Design prestressed concrete sections for flexure using limit state concepts and apply methods such as Magnel's approach.
- Compute short- and long-term deflections in prestressed concrete members and assess crack width control under service loads.
- Analyze statically indeterminate prestressed concrete structures, including secondary moments and concordant cable profiles.
- Understand and design composite structures involving prestressed and in-situ cast concrete under different support conditions.
- Apply engineering knowledge and analytical skills to solve real-world structural design problems involving prestressed concrete.

3. Prescribed Textbook

T1. N Krishna Raju, "Prestressed Concrete", McGraw Hill Education (India) Pvt. Ltd., 6th Edition, 2019.

4. Reference Books

R1. N. Rajagopalan, "Prestressed Concrete", Alpha Science International Ltd., 2nd edition, 2005.

R2. Singh S. B., "Analysis and Design of Prestressed Concrete Structures", Wiley India Pvt Ltd.

R3. IS 1343:2012, Indian Standard Code of Practice for Prestressed Concrete, BIS, 2012.

R4. Lin T.Y., "Design of Prestressed Concrete Structures", Third Edition. John Wiley & Sons. New York, USA

R5. Collins M. P. and Mitchel D., "Prestressed Concrete Structures", Prentice Hall New Delhi.

R6. Naaman A. E., "Prestressed Concrete Analysis & Design Fundamentals", Mc-Graw-Hill.

5. Course Plan

Module Number	Lecture session	Reference	Learning Outcome
1. Introduction, Materials and Prestressing Systems	Introduction and Prestressing systems; Materials used – High Strength Concrete, High strength steel etc.; Prestressing systems –	T1-1,2,3	Understanding the basics of prestressed concrete.



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	pre-tensioning systems, post tensioning systems		
2. Analysis of Prestress and Bending Stress	Basic assumptions and analysis of prestress; Pressure line or thrust line and internal resisting couple; Concept of load balancing; Stresses in tendons; Numerical examples.	T1-4	Understanding the fundamentals of the analysis of prestress and bending stress.
3. Losses of Prestress	Nature of losses of prestress; Reasons for losses of prestress; Numerical examples.	T1-5	Understanding the reasons of prestress losses.
4. Flexural and Shear Strength	Types of failure in flexure; Methods of determining flexural strength for prestressed concrete sections; Numerical examples; Shear resistance of prestressed concrete members; Design of shear reinforcements; Numerical examples.	T1-7,8	Understanding the flexural and shear strengths of prestressed concrete structures.
5. Transfer of prestress in pretensioned members and anchorage zone stresses in post-tensioned members	Transfer of prestress in pre-tensioned members – estimation of the transfer bond stresses, estimation of transverse tensile stresses; Numerical examples; Anchorage zone stresses in post-tensioned members – investigation on anchorage zone stresses; Numerical examples.	T1-9,10	Understanding the transfer of prestress in pretensioned and post-tensioned members.
6. Design of Prestressed Concrete Sections	Design for flexure; Numerical examples; Limitations of Prestress in long span; Limiting zone for the prestressing force; Magnel's method; Numerical examples.	T1-12	Understanding the design of prestresses concrete sections.
7. Deflection of Prestressed Concrete	Factors influencing deflections; Deflection of prestressed	T1-6,11	Understanding the deflections of



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Members and limit state design criteria	concrete members – uncracked and cracked; Numerical examples; Philosophy of limit state design; Crack widths in prestressed members		prestressed concrete members and limit state design criteria.
8. Statically Indeterminate Structures	Statically indeterminate (continuous) prestresses concrete structures; Secondary moments; Equivalent load method; Concordant cable profile and concordant beam; Ultimate strength behavior; Numerical examples.	T1-15	Understanding the statistically indeterminate structures.
9. Composite Construction of Prestressed & In-situ Concrete	Types of composite constructions commonly used; Propped and unpropped constructions; Differential shrinkage; Deflection of composite members; Flexural strength of composite sections; Numerical examples.	T1-14	Understanding the composite construction of prestressed & in-situ concrete

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of components (Close Book / Open Book)
Mid-Semester Test	90 Min.	25	As per the timetable	Close Book
Comprehensive Examination	3 h	40	As per the timetable	Open Notebook (Own handwritten)
Class Tests	-	20	Spread over the semester	Open Book/Open notebook
Assignments	-	15	Spread over the semester	Open Book



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7. Chamber Consultation Hour: To be announced during the lecture.

8. Notices: Notices concerning this course will be displayed on the Department Notice Board.

9. Make-up Policy: If the student is unable to appear for the Regular Test/Examination due to genuine exigencies, the student must refer to the procedure for applying for the Make-up Test/Examination. No Make-up for the Class Tests and assignments.

10. Note (if any):

Academic honesty and integrity are to be maintained by all students throughout the semester, and no type of academic dishonesty is acceptable.

Subhasis Pradhan

Instructor in charge
CE G614